

DSA0502 – QUERY PROCESSING FOR DATA SCIENCE

CLASS PRACTICE PROGRAMS

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1. Write a Python program to validate a password based on the following criteria:

Minimum 8 characters

At least one uppercase letter

At least one lowercase letter

At least one digit

CODE:

```
import re
```

```
password = input("Enter Password: ")
```

```
pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d){8,}$'
```

```
if re.match(pattern, password):
```

```
    print("Valid Password")
```

```
else:
```

```
    print("Invalid Password")
```

OUTPUT:



```
[3]
✓ 11s
import re

password = input("Enter Password: ")

pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*\d){8,}$'

if re.match(pattern, password):
    print("Valid Password")
else:
    print("Invalid Password")

Enter Password: Kirthu$23
Valid Password
```

2. Write a Python program to extract all URLs from a given paragraph using Regular Expressions

CODE:

```
import re
```

```

text = """
Visit https://www.google.com for searching.
Open https://github.com for projects.
Our website is http://example.com.
"""

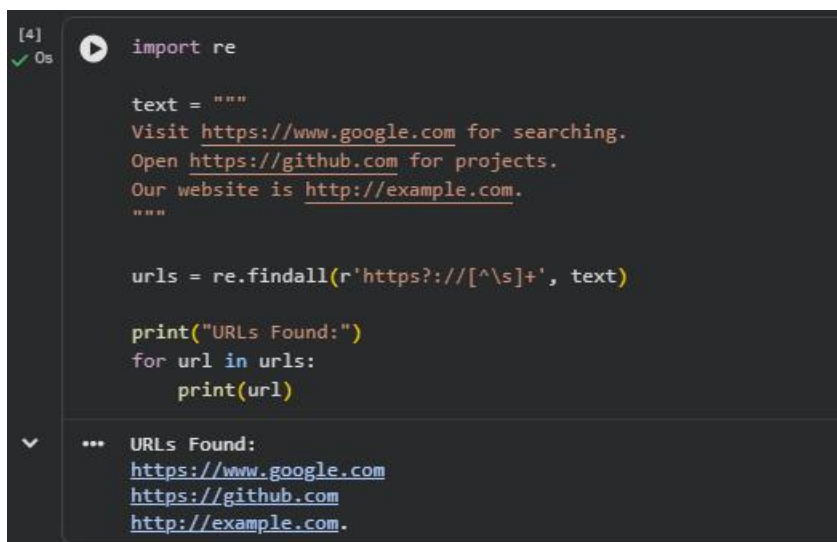
urls = re.findall(r'https?://[^\s]+', text)

print("URLs Found:")

for url in urls:
    print(url)

```

OUTPUT:



```

[4] ✓ 0s
import re

text = """
Visit https://www.google.com for searching.
Open https://github.com for projects.
Our website is http://example.com.
"""

urls = re.findall(r'https?://[^\s]+', text)

print("URLs Found:")
for url in urls:
    print(url)

```

... URLs Found:
<https://www.google.com>
<https://github.com>
<http://example.com>

3. Write a Python program to validate a six-digit Indian PIN code using Regular Expressions.

CODE:

```

import re

pin = input("Enter PIN Code: ")

pattern = r'^[1-9][0-9]{5}$'

if re.match(pattern, pin):
    print("Valid Indian PIN Code")
else:
    print("Invalid PIN Code")

```

OUTPUT:

```
[5]
✓ 5s

import re

pin = input("Enter PIN Code: ")

pattern = r'^[1-9][0-9]{5}$'

if re.match(pattern, pin):
    print("Valid Indian PIN Code")
else:
    print("Invalid PIN Code")

Enter PIN Code: 600125
Valid Indian PIN Code
```

5. Write a Python program to extract all dates in the format DD/MM/YYYY from a given text using Regular Expressions.

CODE:

```
import re
```

```
text = """
```

```
Exam Date: 15/08/2026
```

```
Submission Date: 01/09/2026
```

```
Meeting: 25/12/2026
```

```
"""
```

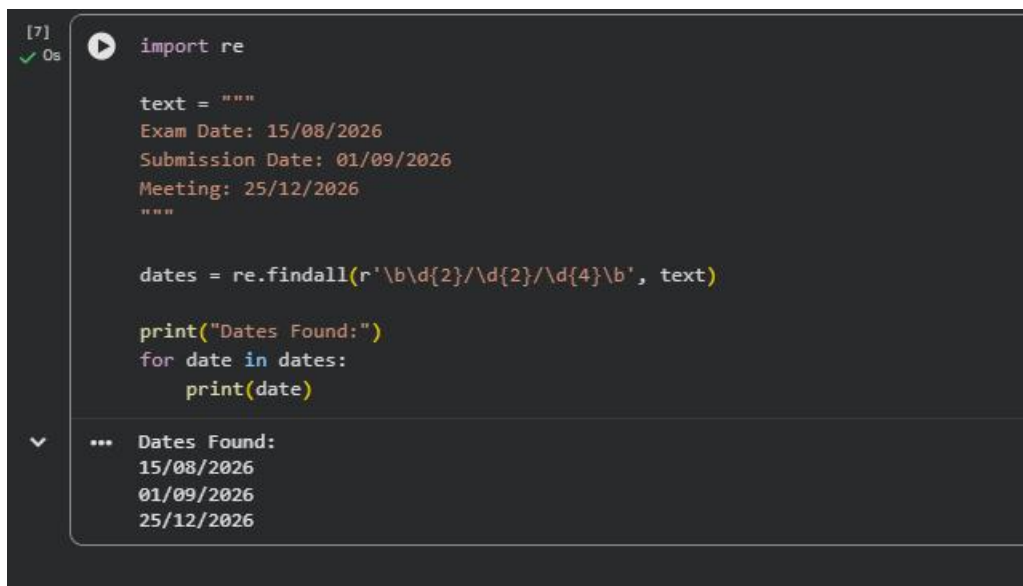
```
dates = re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text)
```

```
print("Dates Found:")
```

```
for date in dates:
```

```
    print(date)
```

OUTPUT:



The screenshot shows a Jupyter Notebook cell with the following code and output:

```
[7] ✓ 0s ▶ import re

text = """
Exam Date: 15/08/2026
Submission Date: 01/09/2026
Meeting: 25/12/2026
"""

dates = re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text)

print("Dates Found:")
for date in dates:
    print(date)
```

▼ ... Dates Found:
15/08/2026
01/09/2026
25/12/2026